NVLink and NVSwitch

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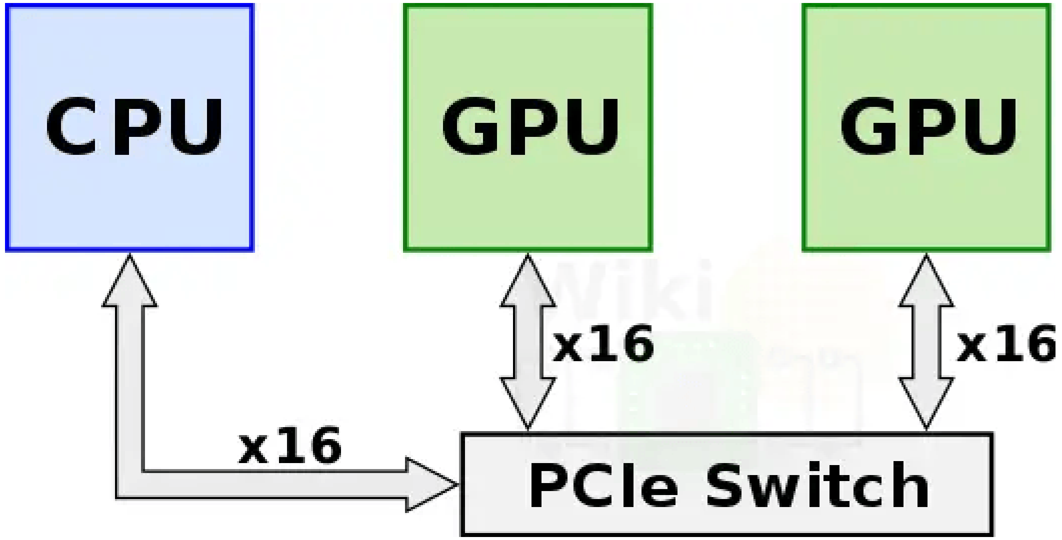
Motivation: Limitations of PCIe

Earlier, multiple GPUs were connected over a PCIe switch and directly to the CPU.

x16 PCIe Gen 3.0 provided only ~32 GB/s bidirectional bandwidth

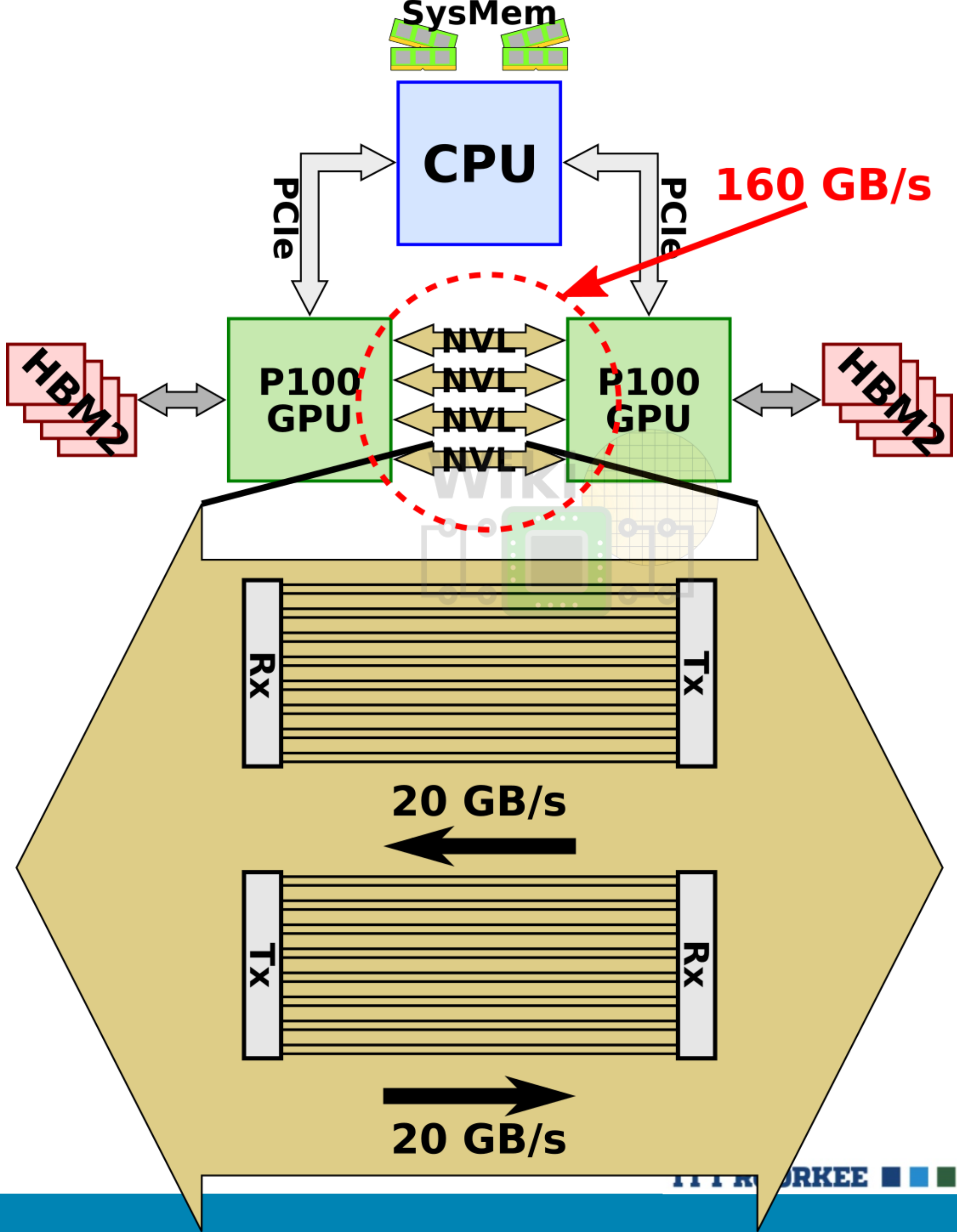


NVLink interconnect

NVLink 1.0, was introduced with the P100 GPU.

P100 incorporated four NVLinks.

With 40 GB/s bidirectional bandwidth per link, the chip had an aggregated bandwidth of 160 GB/s.



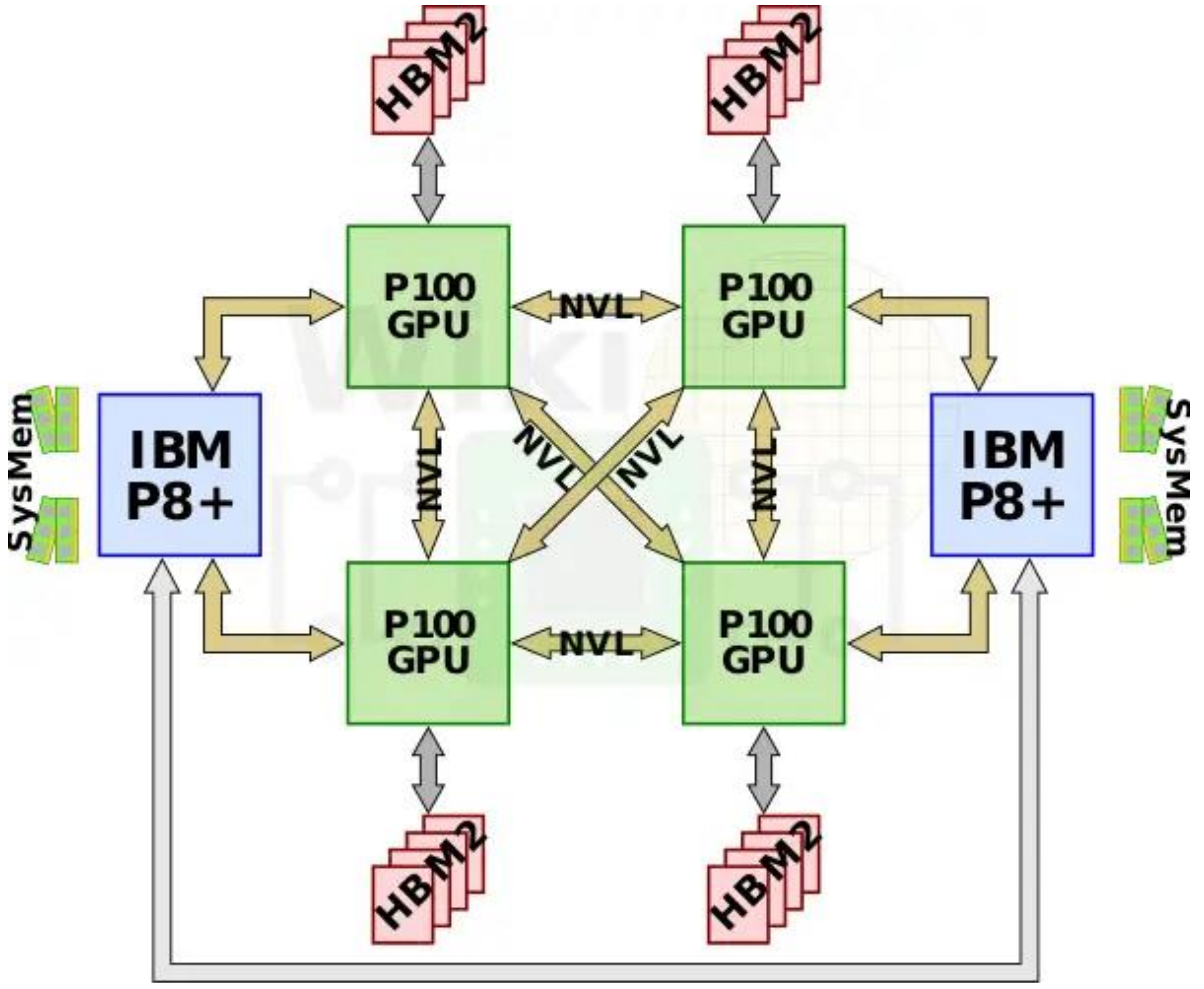
NVLink for connecting to CPUs

NVLink isn't exclusively a GPU-GPU interconnect.

IBM added NVLink 1.0 support to their POWER processors with the POWER8+ microarchitecture.

This allows the P100 GPUs to communicate directly with the CPU over an NVLink instead of PCIe.

A 4-GPU node can be configured in a fully-connected mesh as well as to the nearest POWER8+ CPU.



NVLink in Blackwell GPU

The per-link unidirectional bandwidth in this GPU is 50GB/s, with 18 links

This results in a total bidirectional interconnect bandwidth of 1.8TB/s.

NVLink and NVSwitch

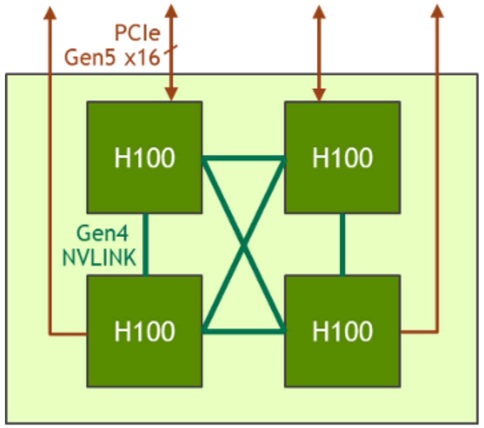
NVLink and NVSwitch are NVIDIA technologies for high-speed GPU communication.

NVLink is a direct, high-bandwidth, low-latency point-to-point interconnect that connects a small number of GPUs, bypassing the slower PCIe bus.

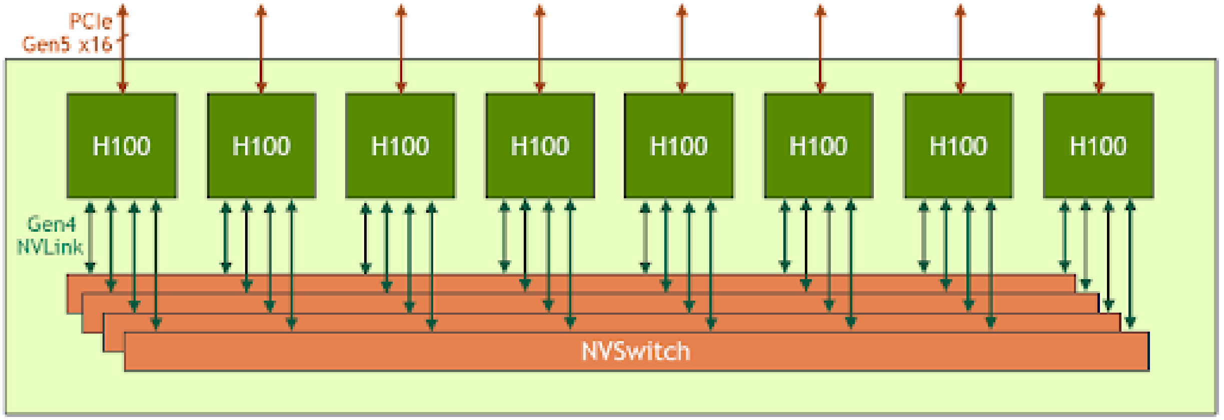
NVSwitch is a high-bandwidth switching fabric that connects multiple GPUs in a single node or system, allowing all GPUs to communicate with each other simultaneously at full speed.

For a single system with many GPUs, NVLink cables physically connect each GPU to a set of NVSwitch chips, and NVSwitch then handles the complex routing to allow any GPU to talk to any other GPU in the system

Feature	NVLink	NVSwitch
Function	Direct GPU-to-GPU connection	Connects multiple GPUs in a system
Scale	Best for a small number of GPUs	Ideal for large-scale multi-GPU systems
Connectivity	Point-to-point, bypassing PCIe	Acts as a central hub for all-to-all communication
Use Case	Low-latency workloads, small-scale AI training, rendering	Enterprise AI, large language models (LLMs), and HPC requiring massive scalability

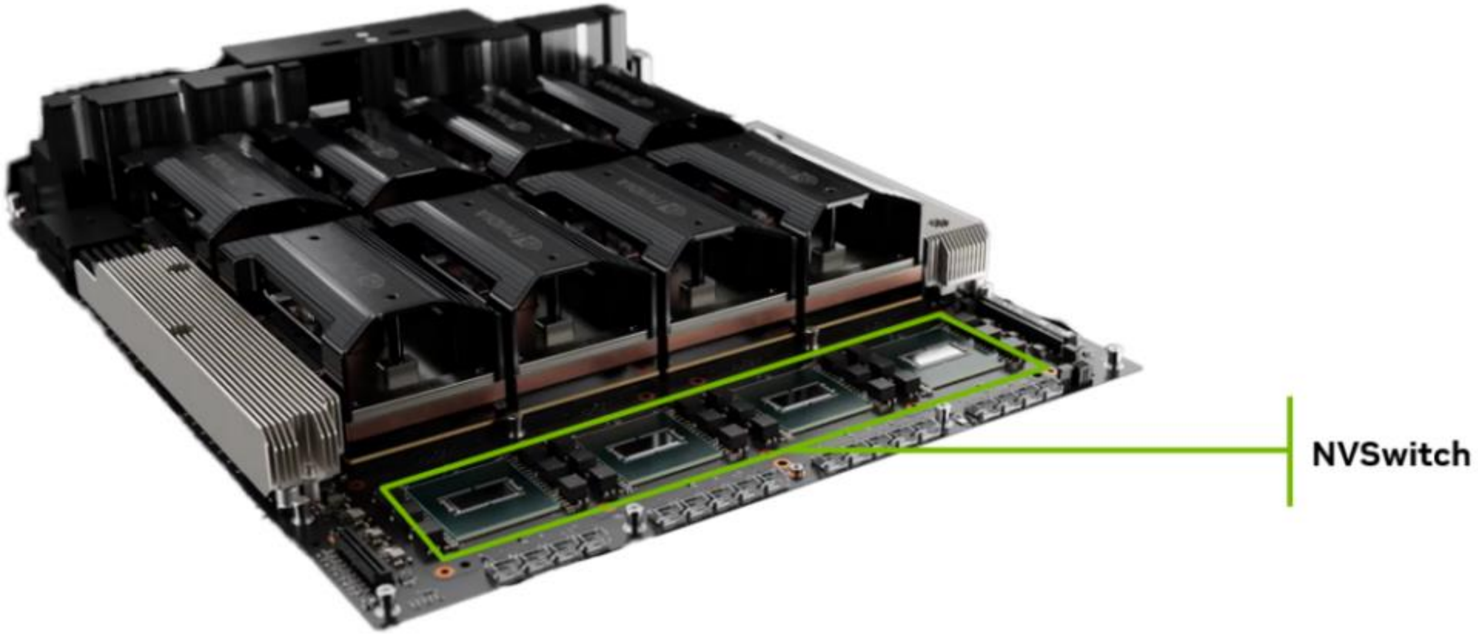


HGX H100 4-GPU: connected with **NVLink**

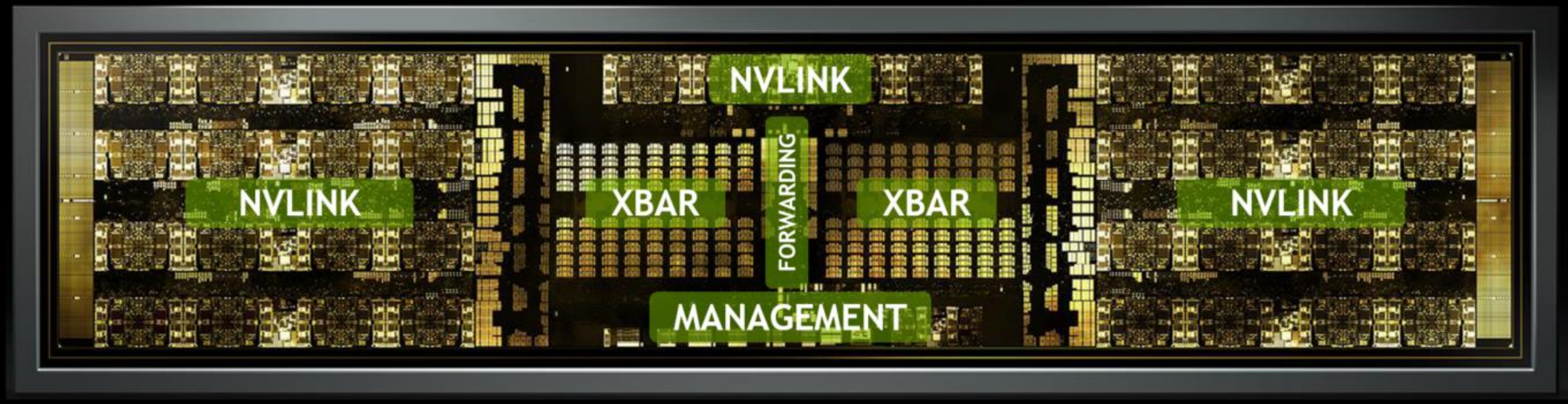


HGX H100 8-GPU: Each H100 GPU has multiple 4th generation NVLink ports and connects to all four **NVSwitches**.

HGX H200 8-GPU with four NVIDIA NVSwitch devices



NVSwitch Die Photo



Server with NVSwitch

NVIDIA Hopper GPU can communicate at 900 GB/s with fourth-generation NVLink.

With the NVSwitch, every NVIDIA Hopper GPU in a server can communicate at 900 GB/s with any other NVIDIA Hopper GPU simultaneously.

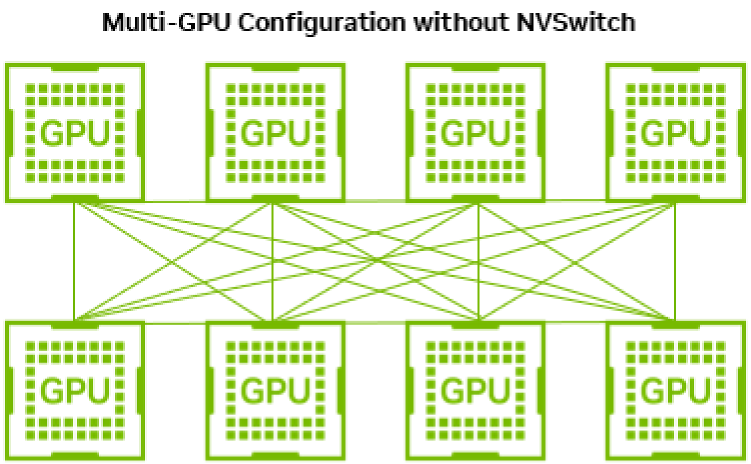
The peak rate does not depend on the number of GPUs that are communicating.

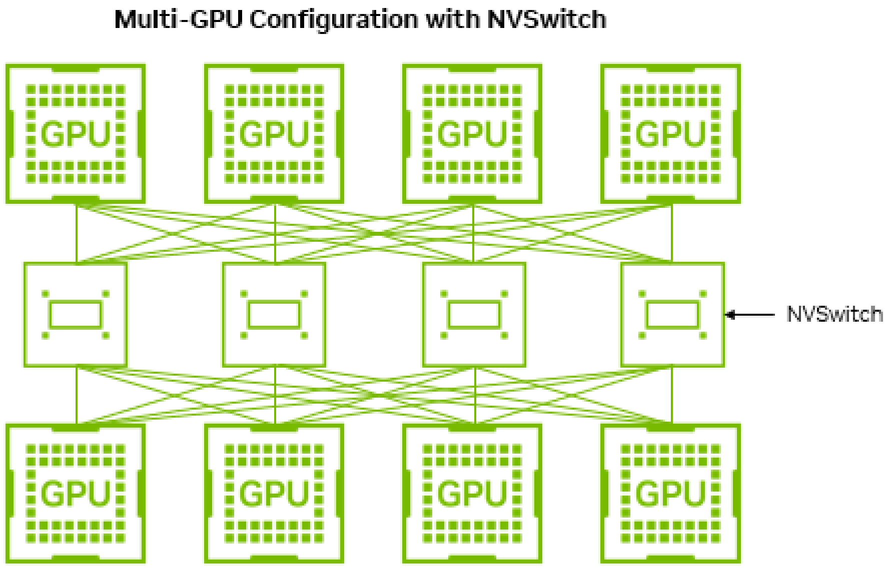
A server without NVSwitch

Consider a hypothetical server with eight H200 GPUs **without NVSwitch** that instead uses point-to-point connections on the server motherboard.

Though it is a lower system cost, each GPU must split the same 900 GB/s connectivity into seven dedicated 128 GB/s point-to-point connections, each connecting to one of the other GPUs in the system.

➔ The speed at which GPUs can communicate depends on the number of GPUs that are communicating.





GPU-to-GPU bandwidth comparison

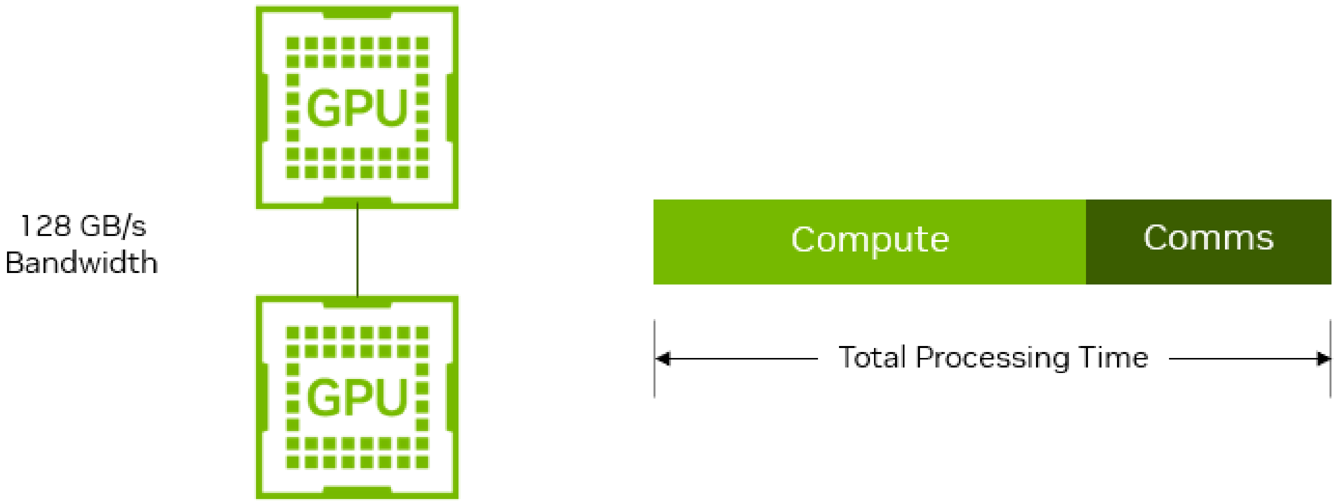
GPU Count	Point-to-Point Bandwidth	NVSwitch Bandwidth
2	128 GB/s	900 GB/s
4	3 x 128 GB/s	900 GB/s
8	7 x 128 GB/s	900 GB/s

Assume 2 GPUs. We want to transfer 20GB of data.

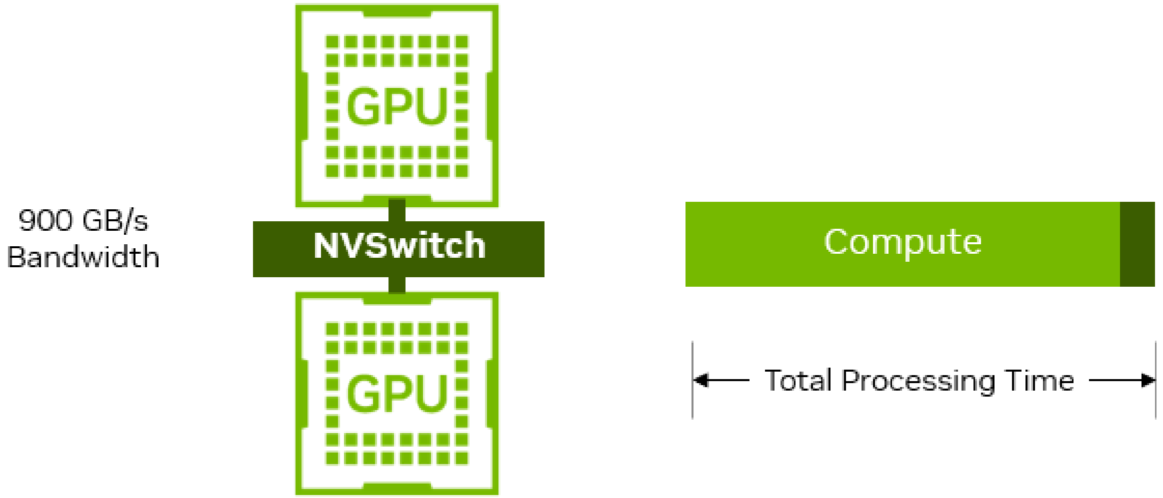
P-2-P takes 20000/128 ms = 156ms

NVSwitch takes 20000/900 ms = 22ms

Without NVSwitch



NVSwitch



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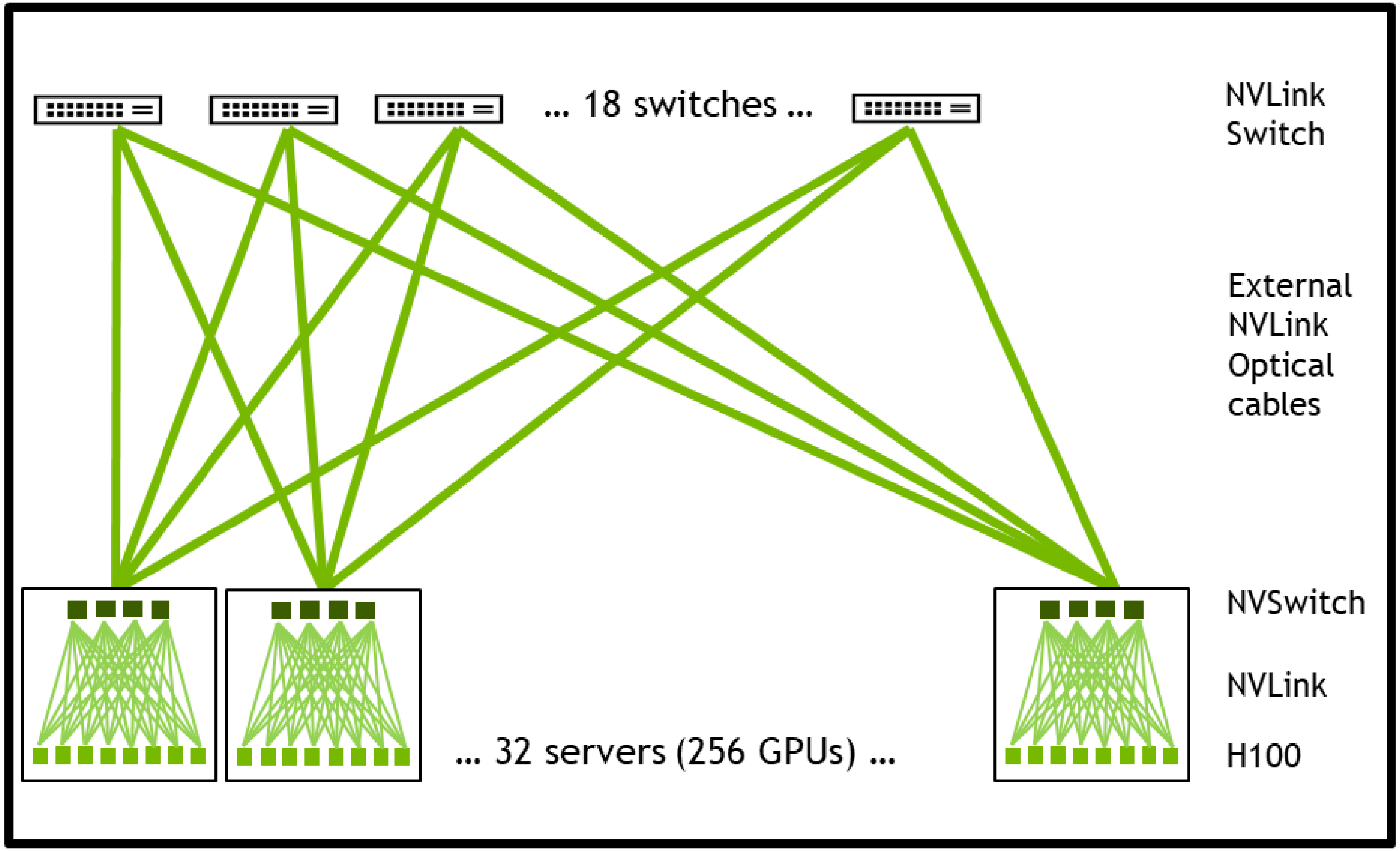
NVSwitch benefit for Llama 3.1 70B inference at various fixed-batch sizes

Batch Size	Throughput tok/s/GPU		NVSwitch Benefit
	Point-to-Point	NVSwitch	
1	25	26	1.0x
2	44	47	1.1x
4	66	76	1.2x
8	87	110	1.3x
16	103	142	1.4x
32	112	168	1.5x

As batch size increases, GPU-to-GPU traffic increases, as does the benefit provided by NVSwitch compared to a point-to-point topology.

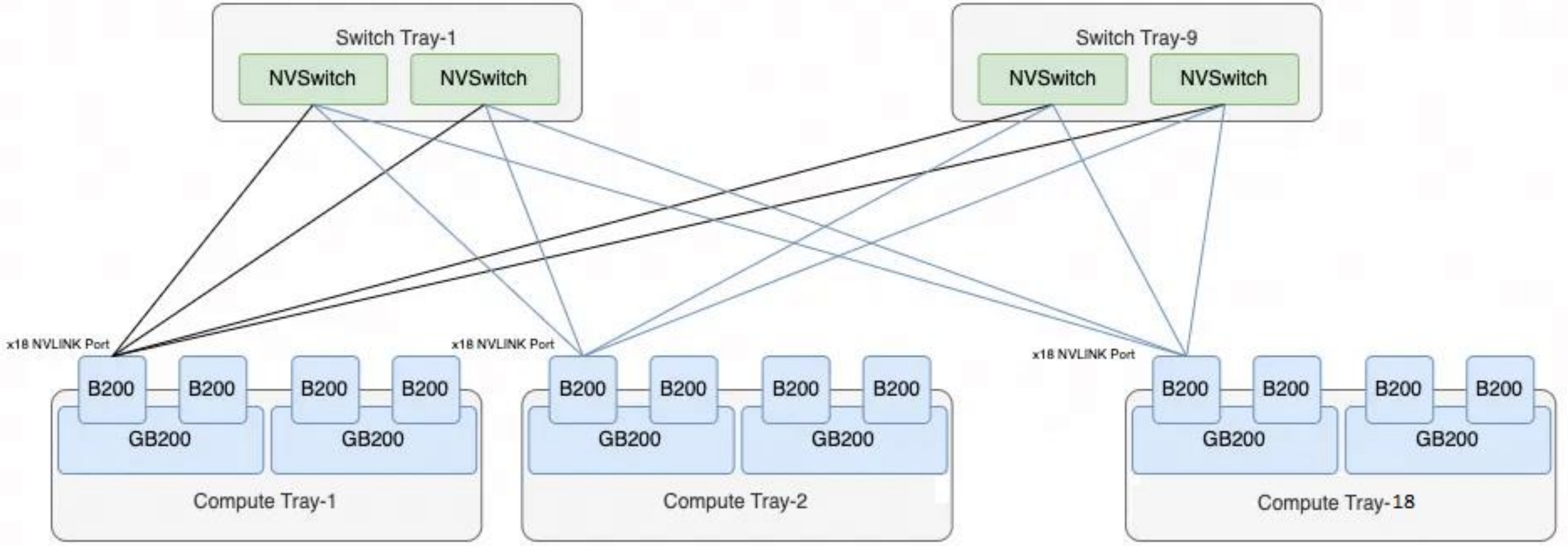
The NVIDIA Blackwell architecture features fifth-generation NVLink, which doubles per-GPU NVLink speeds to 1,800 GB/s.

A Pod with 256 H100 GPUs



NVL72: With 72 GPUs

Nvidia introduced the NVL72, an integrated GPU system that utilizes the NVLink network Switch to achieve full interconnectivity among 72 GPUs.



Each B200 GPU has 18 NVLink ports, resulting in a total of 1,296 NVLink connections (72×18).
A single Switch Tray contains two NVLink Switch chips, each providing 72 interfaces (144 total).
Thus, 9 Switch Trays are required to interconnect the 72 GPUs fully.

Feature		NVIDIA H100	NVIDIA A100	NVIDIA V100
Architecture	Blackwell	Hopper (2022)	Ampere (2020)	Volta (2017)
NVLink Version	NVLink 5.0 (1800 GB/s)	NVLink 4.0 (900 GB/s)	NVLink 3.0 (600 GB/s)	NVLink 2.0 (300 GB/s)
NVLinks per GPU	18	18	12	6
[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
Performance	2.5x H100	3x A100 (FP8 precision)	2x V100	Baseline
Memory	192 GB HBM3	141 GB HBM3	80 GB HBM2e	32 GB HBM2

NVIDIA Blackwell provides maximum flexibility with support for FP64, FP32/TF32, FP16/BF16, INT8/FP8, FP6, and FP4 data formats.

References

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